

the book value, it means wealth has been generated. We can see that if the ratio of market value to book value is equal to one, all three ratios, namely, earnings price ratio (reciprocal of price-earning ratio), yield and return to equity will be equal to each other.

Nobel laureate James Tobin has proposed a valuation ratio, called Tobin's q, after the letter he used to denote it. The q ratio is defined as:

$$\frac{\text{Market value of equities and liabilities}}{\text{estimated replacement cost of assets}}$$

The q ratio looks a little like the market value to book value ratio. But there are two important differences: the numerator of the q ratio includes liabilities of the issuer of debt instruments along with the market value of equities, that is, it includes all wealth. Also, in the denominator of the q ratio, the assets are valued at their replacement cost, not book value.

The main functions of the secondary markets are that it provides useful information both to the issuer as well as to investors, the function it serves for the issuer is as follows. The secondary market provides regular information, by the latest prices of shares, to the original issuer of the shares about the value of its shares. The periodic trading of the asset reveals to the issuer the price the share commands in the market at a certain time. Such information helps the issuer to evaluate how well they are using the funds from earlier primary market activities, and it also provides some idea about how receptive investors will be to new issues.

To the original buyer of shares in the initial public offering in the primary market, the secondary market acts like an escape clause. If the shares do not seem to be doing as well as the original buyer thought, he or she can always sell them for cash. Of course, the investor can also sell it for cash to make capital gains. The secondary market provides the original buyer of the shares an opportunity to sell these for cash. If the investors are not confident that they can move from one financial asset to another, they might be reluctant to invest in any security. This will make it difficult for the issuers to issue new shares, and they will have to offer higher returns or dividends to attract investors.

The secondary markets also perform a useful function. They provide information about the assets' fair price, as well as liquidity from their assets should they want to sell the assets. Another very important function performed by secondary stock markets is that they reduce search costs, by making it easier for buyers and sellers of assets to find each other. The secondary markets also lower transaction costs.

Let us now discuss some basic parameters that enable one to gauge the functioning of the stock market as a whole, which makes it easier to know if the stock market is developed and mature or not. The parameters that we should look at to judge the functioning of equity markets are volatility, liquidity, size of the market, and transaction cost. **The volatility** of a stock is

a measure of the frequency with which its price changes over some time. If the price changes frequently, it means that the stock is volatile. Investors tend to avoid shares with high volatility because of the risks involved. The volatility of stocks also influences the outside flow of funds and hence has a macroeconomic dimension. Proximate factors like speculation and trading and settlement systems, as well as other factors like government policies, interest rates, inflation, and the regulatory framework for equity markets all affect volatility in the stock market.

Liquidity is a very important indicator of the proper functioning of stock markets. The situation about liquidity greatly influences stock market efficiency and development. A market is considered liquid when large volumes of trade can take place without a significant effect on stock prices. If an investor can transact at a price close to the current market price, the market is considered very liquid. There are two methods for measuring liquidity: the turnover ratio and the value traded ratio. The **turnover ratio** equals the ratio of the total value of domestic shares traded to total market capitalisation. This ratio seeks to measure the trading of domestic equities in domestic markets relative to the size of the market. The more the volume of sales, the higher the turnover. High turnover is often an indicator of low transaction costs. Even if a market is large, turnover may be low despite huge market capitalisation if the volume of trading is low, which can occur if the market is not very active. The other measure of liquidity is the **value-traded ratio**. This equals the ratio of the total value of domestic shares traded on the major stock exchanges to the GDP. It is a reflection of organised trading of shares as a proportion of GDP and is thus a macro-level indicator of liquidity in the capital markets.

The third parameter regarding the functioning of equity markets is the size of the stock market, and a useful measure of this size is the **market capitalisation ratio**, which is the ratio of market capitalisation (the value of listed shares i.e., market price $\times$ number of the equity shares) to the GDP. It is economically very significant as it is based on the assumption that the size of the stock market is positively correlated with the ability to mobilise capital and diversify risk. Hence it is expected that in an economy with well-developed stock markets, the market capitalisation ratio will be large. This ratio complements the value-traded ratio. The final indicator of the development, maturity and vibrancy of stock markets is the level of transaction costs. In the context of stock markets transaction costs would include explicit costs like brokerage costs, stamp duty, commission etc., as well as implicit costs like clearing and settlement costs, bad delivery, paperwork and so on. If transaction costs are low, investors will be induced to trade in greater volume, which in turn will raise the value traded ratio.

11.6 INDICES OF SHARE PRICES

The stock market index is a very important indicator of the performance of share prices. It takes a set of stocks as representative of the prices of shares in

the market, and by tracking the changes in this index, we can get an idea of the overall market sentiment. The stock market index is a pointer to market behaviour. The stock market index reflects market direction and indicates day-to-day fluctuations in stock prices.

What are the characteristics of a good index of share prices? A good index should have two characteristics. First, it should consider a set or group of scrips that have high market capitalisation, secondly, these scrips should also be highly liquid. Market capitalisation, as we have seen, refers to the sum of the market value of all the stocks included in the index. The market value is obtained by multiplying the price of the shares by the number of shares outstanding. Liquidity refers to the ease with which a share can be bought or sold at a price close to the current market price.

The share price index is calculated for a particular day as a percentage of the aggregate market value on that day of the set of stocks included in the index to the average market value of the same shares during the base period, for example, the BSE Sensex (sensitive Index) is a weighted average of the prices of 30 select stocks. Suppose the index is the set of prices of the largest 25 companies, it means that these companies are largest in terms of market capitalisation. Sometimes the index is weighted. If the index is based on the price of, say 35 companies, a weighted index implies that a 1% change in the price of the largest companies in the group constituting the index will have more impact on the index than a 1% change in the price of the smallest. Since share prices are changing frequently, the companies with the top shares do not remain the same and hence the constituents of the index can change.

We must be careful to make a distinction between price averages and price indices. In 1884, for example, Charles Dow, publisher of the *World Street Journal* started publishing share price *averages*. The Dow Jones average started with an average of twelve shares. Since 1928, the number of shares whose price average is calculated has been thirty. The main index in Japan, the Nikkei Dow 225, an index of 225 shares is based on average prices, not market capitalisation. Usually, the average is an arithmetic mean, but sometimes geometric mean is calculated. For example, the *Financial Times* Ordinary Share Index which began in 1935, calculated the geometric mean of the prices of 30 shares. Modern indices are based on taking the number of shares and multiplying it by the price. This gives accurate weight to the companies with the largest market capitalisation.

The majority of the stock market indices used in practice fall into two categories:

(i) equal-weighted index, and (ii) value-weighted index. The former is an index that reflects the simple arithmetic average of the sample shares at a certain time (year, month or day) concerning the base period. The latter reflects the aggregate market capitalisation of the sample shares at a certain time concerning the base period. Thus, what we had earlier referred to as

price average can also be viewed as equal-weighted index and what we had referred simply to price index can be called value-weighted index.

Some analysts make a fine distinction even between the market capitalisation-weighted method and the non-capitalisation-weighted method. In this classification, the market capitalisation weighted method can be sub-classified into three types: **full market capitalisation method; free-float market capitalisation method and modified capitalisation method.** In the first, the number of shares outstanding multiplied by the market price of a company's share determines the scrip's weight in the index. The shares with higher market capitalisation would have greater weightage in this kind of index and will exercise greater influence on the overall index. In India, as has been mentioned earlier, the BSE Sensex follows this method. So does the Nifty. The free-float method does not consider a set constructed from among all shares in the market but from a smaller subset of shares. It considers only those shares that are freely available for buying in the market and excludes strategic investments in a company and shares locked in under employee stock-ownership plans. **The free-float method** recognises that investible market capitalisation may be lower than the total capitalisation. In this method, closely held companies would have a much lower weightage. The free-floating methodology is quite popular across the world. The third method under the market capitalisation method is the **modified capitalisation method.** This method recognises that if a simple market capitalisation method is followed, the largest stocks with huge market capitalisation would have a disproportionate weightage and would dominate the entire index. Hence, the modified capitalisation method seeks to limit the influence of the largest stocks in the index.

The two other methods for calculating stock price indices are the **price-weighted index** and the **equal-weighting method.** In the former, the price of each stock in the index is summed up which is then related to a base value. In the equal weighting method, each stock's percentage weight in the index is equal.

Let us now mention some of the major indices of share prices in India and abroad. In India, there are two main indices: the Bombay Stock Exchange (BSE) Sensitive index (Sensex) and the National Stock Exchange (NSE) Nifty. The BSE Sensex (sometimes called S&P BSE Sensex) was launched in 1986. It is a free-float market-weighted stock market index of 30 well-established and financially sound companies listed on the Bombay Stock Exchange. It uses large market capitalisation.

The other major share price index in India is the Nifty of the National Stock Exchange. The National Stock Exchange started its trading in capital market segment in 1994 and in a very quick time, its volume exceeded that of BSE. With collaboration with the Credit Rating Information Services of India Limited (CRISIL) and Standard & Poor, the S&P CNX Nifty was launched on July 8, 1996, comprising 50 scrips, which are selected based on low impact cost, high liquidity and market capitalisation. Both BSE and NSE use

a weighted average method of weighting, where the weights are in proportion to its market capitalisation.

Other than these two major indices, some other indices are the **Economic Times Ordinary Share Price Index, Financial Express Ordinary Price Index, and the RBI Index of Security Prices**. Regional stock exchanges such as those in Ahmedabad, Chennai, Delhi and Kolkata have their indices.

We now state some major indices abroad. A very prominent index is the Dow Jones Industrial Average, which is for the New York Stock Exchange, the National Association of Securities Dealers Automatic Quotation (NASDAQ), the Nikkei in Japan, the Standard & Poor etc.

Check Your Progress 3

- 1) What is the difference between issue capital and paid-up capital? What do you understand by equity premium.?

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- 2) What are the functions performed by the secondary stock markets?

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11.7 LET US SUM UP

In this unit, light has been thrown on financial markets for long-term funds. We have discussed both debt and equity markets. We started by learning about the various participants in the capital market. We also came to know the important functions that capital markets perform. They perform important roles in providing information, enabling matching of sellers and buyers, and lowering search costs and transaction costs. The unit then went on to look at the comparative methods of raising finance through debt and equity. We studied this choice in the context of a representative business firm. This choice, which in general terms is called a firm's capital structure, is part of the firm's financial management.

We next turned to a discussion of debt markets and equity markets and took up their discussion one at a time, separately. We started with the debt markets looked at the instruments traded in that market, and discussed some theories regarding their pricing. We next studied equity markets, distinguishing between primary and secondary markets. We learnt about the way the issues are offered in the primary markets and we also looked at the functioning of

secondary markets and learnt about their functions and uses. We studied in some detail the way to know if stock markets are advanced and mature in an economic system, or are they merely fledgling and poorly developed. We then learnt a bit about the volatility that characterises equity markets and how these are tackled. Finally, indices of share prices were discussed. We looked at the requirements of a good stock price index and came to know the various types of indices in use. The unit finally mentioned some of the important indices in India and abroad.

11.8 KEY WORDS

Coupon	: The periodic interest paid on a bond. It equals the face value of the bond multiplied by the coupon rate of interest.
Coupon Rate	: Annual coupon amount divided by the face value of the bond.
Equity Premium	: When the issue price of an equity share is greater than the par value, the difference is called the equity premium.
Leverage	: The ability of a firm to service its debt.
Market capitalisation	: The sum of the market value of all the stocks included in the index.
Working capital	: Current Assets minus Current Liabilities.
Par Value	: Specified principal sum a bondholder is scheduled to receive when the bond matures.

11.9 SOME USEFUL BOOKS

Bhole, L.M. (2004), *Financial Institutions and Markets*, 4th edition, Tata McGraw-Hill, New Delhi.

Dothan, U. (1990), *Prices in Financial Markets*, Oxford University Press, New York

Fabozzi, F., Modigliani, F., and Jones, F. (2002) *Capital Markets, Institutions and Instruments* 3rd edition, Prentice-Hall, New Jersey.

11.10 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Mechanism for bringing together buyers and sellers of various types of financial assets, making information processing and dissemination easier, help to develop integration among several sectors of the economy.

- 2) Current ratio, quick or acid-test ratio; debt ratio and the debt-equity ratio.

Check Your Progress 2

- 1) value of the rate of interest that makes the present value of the cash stream received from owning the bond equal to the price of the bond.

$$2) A = \sum_{t=1}^T \frac{X}{(1+r)^t} + \frac{M}{(1+r)^T}$$

Check Your Progress 3

- 1) The amount of capital that a company offers to investors is the issued capital. The part of capital subscribed by investors is called the paid-up capital. The par value is the face value of the share. When the issue price is greater than the par value, the difference is called the equity premium.
- 2) They provide information about the assets' fair price, as well as liquidity from their assets should they want to sell the assets, they reduce search costs, by making it easier for buyers and sellers of assets to find each other. They also lower transactions cost.

UNIT 12 BOND MARKETS

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Meaning of Bond Market
- 12.3 Meaning of Bond and their Classification
 - 12.3.1 Meaning of Bonds
 - 12.3.2 Characteristics of Bonds
 - 12.3.3 Classification of Bonds
- 12.4 Valuation of Bond
- 12.5 Bond Yields
 - 12.5.1 Yield to Maturity
 - 12.5.2 Yield to Call
 - 12.5.3 Current Yield
- 12.6 Credit Rating
- 12.7 Bond Market in India
 - 12.7.1 Bond Market - A Part of The Debt Market in India
 - 12.7.2 Regulators
 - 12.7.3 The Growth of Indian Bond Market
 - 12.7.4 Classification of the Debt Instruments
- 12.8 Let Us Sum Up
- 12.9 Key Words
- 12.10 Some Useful Books
- 12.11 Answers/Hints to Check Your Progress Exercises

12.0 OBJECTIVES

After going through this unit, you will be able to:

- state the concept of the bond market;
- define bonds and categorize them according to their different types;
- evaluate the valuation of bonds across various scenarios, including regular coupon bonds, floating coupon bonds, and perpetual bonds;
- discuss bond yields such as Yield to Maturity, Yield to Call, and Current Yield; and
- elucidate and analyze credit rating,

12.1 INTRODUCTION

Suppose that a company needs ₹ 500 crore for some project and it does not want to approach a bank for a loan, then what are the alternatives available? One of the alternatives is the bond market. A bond market is a place where a company can raise loans by issuing a loan-taking certificate (technically known as a bond). Thus, an investor can purchase and sell bonds in a bond market. The present unit aims to discuss the functioning of a bond market and its product i.e., bonds. After reading this unit, you would be able to calculate the value of a bond and make the decision to sell or buy the bond. You will also be able to understand various terms used in the bond market e.g., Yield to Maturity, Yield to Call, Current Yield, Coupon rate, etc. Since bonds carry default and other types of risk, therefore, bonds with good ratings are preferred. The last section of this unit explains the credit rating.

12.2 MEANING OF BOND MARKET

Bonds are loans that investors make to borrowers' entities in exchange for interest payments (zero, fixed, or floating) made over a defined period.

The bond market serves as a way for the borrowers to raise capital, and for investors to earn interest income on their investments. Alternatively, a bond market (also known as a debt market, fixed-income market, or credit market) is the collective name given to all trades and issues of debt securities. The bond market broadly describes a marketplace where investors buy debt securities that are brought to the market by either governmental entities or corporations.

The bond market can be divided into two main categories: the **primary market** and the **secondary market**. In the primary market, new bonds are issued and sold to investors for the first time. In other words, the primary market is frequently referred to as the new issues market in which transactions strictly occur directly between the bond issuers and the bond buyers. In essence, the primary market yields the creation of brand-new debt securities that have not previously been offered to the public. In the **secondary market**, existing bonds are bought and sold among investors. The secondary market provides liquidity to investors, allowing them to sell their bonds before maturity if they need to raise cash or want to invest in other assets. In the secondary market, securities that have already been sold in the primary market are then bought and sold at later dates. Investors can purchase these bonds from a broker, who acts as an intermediary between the buying and selling parties. These secondary market issues may be packaged in the form of pension funds, mutual funds, and life insurance policies — among many other product structures.

The bond market is important to the broader economy because it helps to set interest rates for other financial products, such as mortgages and loans. It also plays a significant role in determining the cost of borrowing for corporations